TECHNICAL REVIEW / TEAM DISCUSSION / 16 OCTOBER 2026

Technical review and Q1 publication plan

Decoupling Compression and Routing in Attention for Time-Series Forecasting

> **Recommended primary target: *Neurocomputing*. Strengthen the paper as an explanatory, controlled study of when compression and routing are useful, rather than promote DRRA (8,4) as a generally superior configuration.**

The draft contains a useful factorial study and substantial reported audit work. Its present vulnerability is not editorial quality: several comparisons do not yet isolate the proposed explanation, and the practical efficiency claim is undermined by the manuscript's own last-query measurements. The priorities below address those issues before expanding the benchmark suite. (Manuscript, pp. 5–8, 14 and 24.)

Journal	Verified ranking basis	Recommended role
Neurocomputing	2025 SCImago/SJR: **best quartile Q1**	Primary target for a rigorous design study with meaningful forecasting and systems evidence.
Neural Networks	2025 SCImago/SJR: **best quartile Q1**	Stretch target after a stronger mechanism-level contribution and convincing transfer.
Information Sciences	2025 SCImago/SJR: **best quartile Q1**	Alternative for a broadly useful compression/selection analysis and validated decision framework.

Ranking sources: SCImago (n.d.-c), SCImago (n.d.-b) and SCImago (n.d.-a). Journal positioning is my assessment of fit, not an acceptance prediction.

What a stronger submission should establish

Validity: correct the interaction interpretation, match projection architecture, audit the selected information budget, and separate untrained indexer parameters from changing route decisions. **Practical relevance:** use the computation actually required for forecasting and demonstrate nondegenerate routing in a competent temporal backbone. **Reusable insight:** explain where compression helps, where routing adds value, and where simpler alternatives suffice.

Ranking and review scope

Here, Q1 means the **2025 SCImago/SJR best quartile, based on Scopus data**. It is not a claim about a particular CiteScore category percentile or a JCR quartile. Direct Scopus source-detail pages were inaccessible; a category-specific CiteScore requirement must therefore be checked through the team's institutional access before submission. (SCImago, n.d.-d) (Elsevier, n.d.-a)

Basis: the supplied 27-page NewTransformer.pdf. Page references identify that draft. Numerical differences below are recalculated from printed tables; code, checkpoints and raw manifests have not been inspected or executed. Implementation-dependent concerns are explicitly conditional. Priorities are recommended evidence standards, not formal journal rules.

P0 / CORRECTIONS REQUIRED BEFORE NEW HEADLINE RESULTS

1. Correct the statistical interpretation and baseline attribution

R1. Interpret the interaction using simple effects

Section 4.2 says coarsening KV compression helps more when routing is fine. Table 2 shows the reverse. Using its validation means, the changes below follow directly; negative values indicate lower MSE. (Manuscript, p. 7, Table 2 and §4.2.)

Change	Fixed condition	Δ validation MSE	Reading of the mean
cKV: 4 to 8	Fine routing: cR = 4	+0.0042	Very small deterioration
cKV: 4 to 8	Coarse routing: cR = 8	−0.0682	Improvement
cR: 4 to 8	Fine content: cKV = 4	+0.0662	Deterioration
cR: 4 to 8	Coarse content: cKV = 8	−0.0062	Very small improvement

The interaction is principally associated with the poor (4,8) cell, not a demonstrated universal benefit of fine routing. At cKV = 8, the finer router does not improve the mean validation error. Recompute paired confidence intervals for these simple effects from the unrounded seed data; these printed-mean calculations do not establish their significance.

Required revision: make an interaction plot the main factorial figure, report the conditional effects, and revise the abstract, §5.1 and conclusion accordingly. H1 is algebraically the off-diagonal comparison $y_{4,8} - y_{8,4}$; it is not independent confirmation of two general mechanisms. The cross-dataset reversals should help define the research question, rather than appear only as limitations. (Manuscript, p. 8, §4.2.)

R2. Separate attention savings from projection savings

The dense baseline has a rank-128 output projection; the compressed cells use rank 64. Table 9 assigns these projections 20.97M and 10.49M MACs respectively. Consequently: (Manuscript, p. 24, Table 9.)

Reported total saving = 76.86 − 59.63 = 17.23M MACs
Projection difference = 20.97 − 10.49 = 10.48M MACs
Projection share of the saving = 10.48 / 17.23 ≈ 61%

Replacing only the baseline projection cost gives 66.38M MACs, reducing the accounting advantage to approximately **10.2%, not 22%**. This is an accounting sensitivity calculation, not a prediction of retrained accuracy or measured latency; the remaining operation counts still require audit.

Required experiment: add a mechanism-matched dense baseline with the same projection ranks, normalization, positional treatment and compatible sink treatment. Retain the original Normal model as an as-run reference. A separately tuned, equal-budget dense model is also useful for deployment comparison. Do not conflate parameter matching with architectural matching, or reference-output agreement with comparison fairness.

> **Submission condition: every headline improvement must state whether it is attributable to compression/routing, changed projections, a different prediction head, or an implementation choice.**

P0 / IDENTIFICATION AND IMPLEMENTATION AUDITS

2. Isolate routing resolution and verify what actually changes

R3. Fixed top-k is not a fixed information budget

At k = 16, one selected routing token in (4,8) expands to two KV blocks, whereas selections in (8,4) can collide on the same KV block. The draft also reports approximately 14 unique selected blocks for the equal-rate reference versus 10 for the dual-rate cell. Thus, the original factorial compares complete configurations; it does not isolate routing resolution independently of realized coverage. (Manuscript, pp. 5–6, §§3.3–3.4.)

Required experiment: retain the fixed-k factorial, then add comparisons that fix the number of *unique*, *valid KV blocks* after mapping and deduplication, at each fixed cKV. Specify how index scores are aggregated to block scores and how the selector fills the budget. Apply identical causality rules and handle early queries with too few eligible blocks explicitly.

Log candidate count, eligible count, unique selections, duplicate rate, selected fraction, lag distribution and the union of raw-time receptive fields. The compressor overlaps adjacent windows, so block count times cKV is not an exact measure of unique temporal coverage. For comparisons across cKV, add a separate coverage-matched sensitivity analysis rather than claim that all budgets can be matched simultaneously.

R4. An untrained indexer is not necessarily a fixed route

The router reuses queries and hidden representations produced by trainable components. Even if indexer-exclusive parameters receive no direct task-loss gradient, the selected blocks can change as those inputs change. The repeated phrase “routing is fixed at initialization” is therefore not established by the reported gradient audit. (Manuscript, p. 5, §3.3; p. 26, Table 10.)

Required audit: track route overlap and entropy on fixed probe inputs at initialization and across epochs. Save parameter snapshots and distinguish a missing gradient from a zero gradient; check actual optimizer updates. Separate backbone, indexer and data-loader random seeds. Compare the present indexer with a genuinely fixed mask and with one explicitly trained selector.

A teacher-guided block-importance loss is a practical learned-routing control, but report its extra training cost and test-time requirements. DeepSeek-V4 explicitly includes an indexer warm-up stage, so its training regime is not interchangeable with the manuscript’s untrained-indexer setting. (DeepSeek-AI, 2026)

R5. Rebuild the evidential chain after the mapping correction

Some mechanism probes are retained from checkpoints predating the mapping correction. They cannot substantiate the corrected method’s mechanism. Regenerate those results and their figures from a single versioned implementation, or remove them from the evidential argument while retaining their development history in the archive. (Manuscript, p. 9, §4.3.)

Extend the reported future-invariance suite to indexer scores, block boundaries, ties, padding and every tested rate pair. Zero future gradients alone cannot exclude future dependence through a hard discrete selector. The existing perturbation-based checks are valuable; the revised report should state their coverage and preserve the passing logs, rather than assume a failure.

P0 / COMPUTATIONAL VALIDITY AND BACKBONE TRANSFER

3. Use the forecast-required path and nondegenerate routing

R6. Make optimized forecasting cost the primary comparison

Equation 1 uses the final hidden state of a single block. The manuscript's last-query audit reports Normal at **2.6–3.9 ms** versus DRRA at **5.8–8.6 ms** per batch-32 pass. That comparison reverses the full-layer efficiency ranking. It must lead the efficiency results and the accuracy–latency frontier. (Manuscript, pp. 4 and 14, §§3.2 and 4.7.)

For the stated one-block, pointwise-readout architecture, the last query can attend to all historical keys without calculating all query outputs. Its attention-core cost is linear in L for fixed width. Verify forecast and parameter-gradient equivalence before using this pruned graph for training; stochastic operations must be handled consistently. Do not generalize the simplification to deeper architectures without tracing their dependencies.

Required revision: report FULL operator timings separately from forecast-required timings, optimize every comparator legitimately, and replace Figure 6 with the deployment-relevant frontier. Full-layer savings may remain useful in an architecture whose head needs all temporal outputs, but cannot support the current per-forecast efficiency headline.

R7. Check whether the PatchTST experiment actually routes

With lookback 256, patch length 16 and stride 8, PatchTST has 31 tokens before optional end padding, approximately 32 with common padding. Compression at cR = 4 or 8 then gives only about eight or four routing candidates. If k = 16 is retained and the module acts along the patch axis, all available candidates are selected: routing resolution is not meaningfully tested. (Manuscript, p. 10, §4.5; p. 23, Table 8.) (Nie et al., 2023)

This is an **implementation-dependent hypothesis, not a confirmed bug**. Report the actual tensor axis, patch padding, candidate count, k and selected fraction. At L = 1024 the same patching yields only about 128 tokens, so cR = 8 with k = 16 may still select everything. Use a smaller budget, such as k = 4 or 8, or a sufficiently longer token sequence, and verify active selection in every cell.

Do not interpret small transfer effects as evidence that a strong backbone washes out routing until saturation is excluded. Include the original dense PatchTST backbone and the no-routing compressed backbone in both accuracy and speed comparisons.

R8. Reconcile the remaining accounting before extrapolation

Table 1 lists 17 softmax columns for (32,4), but its stated C + W + 1 gives 256/32 + 16 + 1 = **25**. Check whether the table is wrong or W changed. Also expose the number hI and dimension dI of scoring heads: the dot-product routing term is generally hI × dI × NQ × (L/cR), where NQ is the number of evaluated queries. Equation 2 assumes a particular head arrangement that requires tensor-shape verification. (Manuscript, pp. 6, 15 and 24.)

Recompute every affected total and scaling curve after these audits. Matching one script to another script using the same assumptions is not an independent validation of the assumptions.

P1 / EXPERIMENTS WITH THE HIGHEST DIAGNOSTIC VALUE

4. Establish whether the router is needed at all

The current dense-masked implementation computes over every compressed column even when most are masked. Removing routing therefore need not widen the attention core; it removes the indexer and selector. This makes all-block compressed attention the decisive missing control. (Manuscript, p. 5, §3.3; p. 26, Appendix F.)

Control	Question answered	Essential matching/reporting
Projection-matched dense	Do gains survive removal of the projection confound?	Match ranks, width, normalization and readout; disclose sink and positional differences.
All-block compressed, no router	Is compression alone sufficient?	Test cKV = 4 and 8; preserve the compressor, local branch and head.
Window-only / no historical branch	Does forecasting need the compressed history?	Keep the same backbone and prediction head; report the local-window size.
Random, recent and evenly spaced block selection	Does content-based selection outperform inexpensive alternatives?	Match unique valid blocks and compare lag/coverage distributions.
Mean or filtered pooling	Is the learned overlapping compressor better than simple smoothing?	Match output block count; separate filter width from routing resolution.
Trained indexer	Does learning block importance change the factorial result?	Report training loss, optimization budget, added parameters and inference overhead.
Original dense forecasting backbone	Is the modified model useful beyond the minimal testbed?	Use a competent temporal-token architecture and include its unmodified version.

Run the controls in a diagnostic sequence

First test projection-matched dense, all-block compression and the four factorial cells on ETTh1 and a contrasting dataset. Then use window-only and inexpensive selection rules to determine whether remote content and content-dependent selection contribute. Only after that should the learned-indexer and broader transfer experiments be expanded.

Use the same fixed training protocol for this mechanism track, but verify convergence and include a budget sensitivity check. Separately retrain ablations and perform checkpoint-level interventions: replacing a trained router only at evaluation measures sensitivity to removal, whereas retraining without it measures whether the remaining architecture can adapt.

> **Decision rule for the team: if all-block compressed attention matches or improves accuracy and optimized latency, do not preserve the router merely to preserve the name DRRA. A well-characterized finding that routing is unnecessary in a defined regime can be the stronger contribution.**

A credible submission need not win every dataset. It should resolve a question that the controls could genuinely answer against the preferred design, and show that the answer remains informative outside the original single-block setting.

P1 / GENERALIZATION AND FAIR FORECASTING COMPARISONS

5. Expand along dimensions that can change the conclusion

Two evaluation tracks are necessary

Mechanism track: hold the temporal backbone, normalization, prediction head and optimization recipe fixed while changing compression, routing and budget controls. **Practical forecasting track:** compare complete systems using validation-selected settings and equal, disclosed tuning budgets. An identical optimizer for unlike architectures is a useful control, but is not sufficient evidence of a fair best-achievable comparison.

DLinear and PatchTST should appear on the actual error–latency–memory frontier, not only as calibration rows. Table 5 reports ETTh2 MSE of 1.8080 for DRRA versus 0.2753 for DLinear; on the change-point generator the values are 6.0013 and 0.0622. These gaps make backbone competence a central issue. Include persistence and, where a defensible period is known, seasonal persistence. (Manuscript, p. 12, Table 5.) (Zeng et al., 2023)

Dimension	Recommended core revision	Reason
Datasets	Start with ETTh1, ETTh2, Electricity and Exchange; use remaining datasets as broader validation.	These already show different factorial behavior in the draft.
Forecast horizon	H = 96, 336 and 720 on a planned subset.	Tests whether rate effects depend on forecast distance.
Lookback	L = 256 and 1024 for raw temporal tokens; choose patch settings by their actual token count.	Tests accuracy and computational cost at the same longer context.
Backbone	Minimal model plus one competent multilayer temporal/patch backbone.	Tests whether the conclusions survive a more relevant architecture.
Replication	Pilot with paired seeds; predefine the confirmatory seed count and temporal holdouts before running.	Avoids expanding runs until a desired significance threshold appears.

Do not run every diagnostic control over the entire Cartesian product. Use the focused controls to identify the explanation, then test the four factorial cells and decisive comparators on the planned generalization matrix. The original 12-dataset suite can provide breadth once implementation and protocol inconsistencies are resolved.

Keep token axes and data protocols explicit

iTransformer is useful as an external forecaster, but its attention operates over variate tokens. A direct module replacement would therefore test compression of variables, not the same temporal-compression hypothesis. Use it as a system comparator unless a separate hypothesis is stated. (Liu et al., 2024)

For literature comparability, add a clearly separated standard-benchmark protocol alongside the frozen 60/20/20 split. Within each track, all methods must use identical forecast origins, target channels, information availability and train-only scaling. Preserve a fresh temporal holdout: previously inspected test periods cannot become unseen merely by renaming the endpoint. TFB offers a useful reference for consistent, diverse evaluation. (Qiu et al., 2024)

P1 / P2 / FROM CONFIGURATION COMPARISONS TO EXPLANATION

6. Build a mechanism-level contribution that others can reuse

Replace suggestive correlations with controlled signal tests

The reported lag-1-autocorrelation association, r approximately −0.3 with p approximately 0.3, is descriptive rather than an established explanation. The periodic delayed-copy generator also allows periodic structure to substitute for arbitrary-content retrieval. (Manuscript, pp. 9–10, §4.3.)

Synthetic family	Vary independently	Measure beyond pooled MSE
Smooth latent signal plus noise	Signal bandwidth, observation noise and compression rate; specify whether the target is noisy or latent.	Predictive-signal distortion, noise attenuation and per-horizon error.
Random-content delayed recall	Fresh episodes, in-window lag and event alignment; provide an observable lag cue when necessary.	Recall error on recoverable channels and selected-block coverage of the target.
Localized predictive events	Event width, lag and block-boundary alignment; avoid targets that require unobserved random content.	Event localization, amplitude error and sensitivity to compression.

These tests distinguish redundancy of the observed series from preservation of the information needed for forecasting. A nonlinear, gated, overlapping compressor should not be called a low-pass filter without measurement; compare it with a specified linear filter and record its empirical response.

A useful analytical starting point

For a checkpoint-level comparison with matched representations, let o_{full} denote uncompressed attention, o_{all} attention to all compressed blocks, and o_{route} routed compressed attention. The triangle inequality gives:

$$||o_{full} - o_{route}|| \le ||o_{full} - o_{all}|| + ||o_{all} - o_{route}||$$

With fixed compressed logits and values, suppose routing discards attention mass τ from the all-block distribution, the retained set is nonempty, and every value norm is bounded by V_{max}. Renormalization then gives:

$$||o_{all} - o_{route}|| \le 2V_{max}\tau$$

To see this, write the all-block output as $(1 - \tau)\mu_{kept} + \tau\mu_{removed}$; its difference from μ_{kept} is $\tau(\mu_{removed} - \mu_{kept})$. Keep local and sink contributions in the accounting. This is an elementary proposed diagnostic, not a novelty claim or a guarantee of forecast accuracy.

Required evidence: measure compression discrepancy and discarded mass, then test whether they predict the observed rate effects. Also measure forecast changes under targeted interventions, because an attention-output bound alone does not establish predictive importance. For a more ambitious Neural Networks submission, develop and validate a genuinely informative analysis beyond this elementary decomposition.

> **Potential reusable result: a held-out-validated selection guide connecting signal structure, token geometry and realized computational budget to when compression alone or compression plus routing is preferable.**

P1 / RELIABLE INFERENCE, PERFORMANCE AND REPRODUCTION

7. Strengthen the statistics and systems evidence

Separate three sources of uncertainty

Training randomness: preserve paired-seed analyses for the fixed dataset. **Temporal uncertainty:** evaluate rolling forecast origins or independent time blocks and retain per-origin losses. **Dataset variation:** report conditional effects by dataset and avoid counting every seed or related ETT series as an independent domain. These answer different questions; one large seed count does not replace the others. (Demšar, 2006, for across-dataset comparison principles)

For per-origin inference, use a paired moving-block or other dependence-aware resampling scheme with block-length sensitivity; do not bootstrap overlapping windows as independent observations. Report test MSE and MAE, uncertainty for the interaction and simple effects, and the multiplicity family for primary contrasts. State whether each interval concerns a paired mean, median or another estimand: a mean-difference interval and a signed-rank test need not agree.

A nonsignificant difference is not evidence of preserved accuracy. To claim that a cheaper configuration retains performance, predefine a defensible non-inferiority margin and estimate whether the data support it. Do not select a convenient margin after seeing the results. Preserve the original five-seed analysis, its extension and the new confirmation as distinguishable stages. (Manuscript, pp. 7–8 and 11–12, §§4.1–4.6.)

Benchmark optimized implementations, not only operation counts

Use the existing Apple CPU measurements as one platform and add a representative GPU or a clearly justified second deployment platform. Compare against the strongest valid dense implementation, including fused attention where compatible. FlashAttention demonstrates why reduced memory traffic and arithmetic counts must be distinguished. (Dao et al., 2022)

Measure forecast-required latency at batch 1 and a realistic batched setting, throughput, peak allocated memory, forward-plus-backward time and end-to-end training time. Report dtype, kernel, compilation mode, thread count, synchronization, warm-up, device and software versions. Alternate timed configurations within rounds and report paired uncertainty; include compression, index scoring, top-k, masks, transfers and the prediction head in the timed path.

The current implementation pays the full compressed attention width despite sparse masks. Report this explicitly and profile where time is spent. A GPU sparse kernel is valuable only if a measured bottleneck justifies developing it; do not require bespoke kernel work merely to enlarge the paper.

Treat energy as a measurement question

Move the present arithmetic-to-energy projections to a qualified appendix, or replace them with sustained, instrumented measurements. For joules per forecast, state whether the boundary is device-only or whole-system, report idle treatment, integrate power over a timed workload, and quantify measurement resolution. MAC reductions alone establish neither the size nor the sign of an energy improvement. (Manuscript, §§4.8 and Appendix D.)

Release a tagged code archive, dataset hashes, configuration files, raw paired results, profiler traces, corrected tests and one-command table/figure regeneration. Distinguish the draft's claims that artifacts exist from an independently verified release; archive the exact version used for submission.

POSITIONING / NOVELTY, FIGURES AND CITEABLE OUTPUTS

8. Reframe the manuscript around a useful conditional finding

Recommended title and contribution

> **When Does Compressed Attention Help Time-Series Forecasting?**
> **Separating Compression, Routing, and Computational Cost**

Keep the two-axis design study as the paper's identity. The revised contribution should be an explanation supported by controlled comparisons and meaningful execution costs, not a claim to have invented the component mechanisms. Replace repeated defensive scope statements with a compact limitations section and explicit boundaries in each claim.

Update the closest-work comparison substantively

MoBA: it varies block granularity while adjusting top-k to maintain constant sparsity. Discuss how your independently compressed content stream differs, and why realized-budget controls matter. (Lu et al., 2025)
Native Sparse Attention: it distinguishes compression and selection blocks and maps importance between different geometries; do not claim that separating these granularities is itself unprecedented. (Yuan et al., 2025)

The Sparse Frontier: its revised analysis emphasizes task- and execution-stage-dependent trade-offs. Your opportunity is a forecasting-specific explanatory study, not simply another sparse-attention leaderboard. (Nawrot et al., 2026) Identify the (4,4) cell as an equal-rate configuration, not a full DeepSeek reproduction; state training, window, score and implementation differences explicitly.

Replace the principal figures, not merely their captions

Proposed figure	Scientific purpose
1. Execution geometry	Token axes, candidate/selected blocks, and FULL versus forecast-required paths.
2. Interaction effects	Paired conditional effects and uncertainty across datasets, not just ETTh1 main effects.
3. Matched controls	Residual benefits after matching projections, coverage and training conditions.
4. Mechanistic map	Controlled signal properties, attention diagnostics, and validated successes and failures.
5. Accuracy–resource frontier	Dense, compression-only, routed and external models; uncertainty and actual execution costs.

The principal results should follow that same logic: identification, mechanism, generalization and practical value. Move development history, stale experiments, unrelated surrogate work and purely extrapolated energy calculations out of the central narrative. Do not remove negative results that remain scientifically informative after correction.

For reuse and citation potential: release the controls and synthetic tests as a documented benchmark. Validate a selection guide on held-out cases and report its regret relative to the best tested configuration, including failures.

TEAM DELIVERY PLAN / PRIORITIES AND SUBMISSION GATES

9. Execute in stages and let the results determine the final claim

Stage / suggested lead	Required work	Completion evidence
A. Validity Methods + implementation	R1–R8: conditional effects, matched projection, routing budget, parameter/route audit, corrected provenance, last-query graph, patch geometry and counts.	Versioned audit report; corrected tables; no unresolved headline-confounding issue.
B. Explanation Modelling + analysis	Decisive ablations, signal-controlled probes and checkpoint interventions.	Evidence distinguishing compression, selection and local-history effects.
C. Relevance Benchmark + systems	Competent backbone; horizon/lookback extension; external systems; temporal validation; optimized multi-platform timing.	Held-out accuracy–latency–memory frontiers and clearly delimited regimes.
D. Submission Lead authors	Revise claims and figures; archive code/data; complete journal-metric and source checks.	Traceable manuscript, supplement and reproducibility package for the chosen venue.

Minimum package I recommend for Neurocomputing

Resolve all P0 issues; include projection-matched dense and compression-only controls; demonstrate active routing in a competent backbone; add targeted multi-horizon/longer-lookback and temporal validation; and make the main computational claims survive optimized forecast-required accounting. A trained-router control and the mechanistic tests should support the explanation rather than remain general future-work promises.

The submission should establish at least one meaningful, reproducible conclusion: a genuine routed advantage in a defined regime; a useful compression-only trade-off; or an explanatory boundary showing when routing is unnecessary or harmful. Uniform superiority is not a requirement of this recommendation. Unresolved confounds cannot be compensated for by more datasets or more cautious wording.

What would justify choosing Neural Networks instead?

Choose the stretch target when the revision delivers a stronger mechanism-level result or predictive theory, validates it across competent backbones, and demonstrates that it is not explained by projection rank, budget, saturation or optimization artifacts. A larger benchmark table alone would not change my target recommendation. Information Sciences becomes attractive if the principal output is a generalizable compression/selection framework rather than an architecture-specific case study.

Claims to remove unless the new evidence supports them

Do not retain “fine routing carries accuracy” as a general rule; “fixed routing” based only on gradient norms; the 22% whole-model MAC saving as an attention-only benefit; “accuracy preserved” based on a nonsignificant test; “backbone washes out effects” without a saturation audit; or deployment/energy superiority inferred from full-layer arithmetic. Each should be replaced by the specific comparison and operating conditions actually established.

The staged plan is a research recommendation, not an assurance of Q1 acceptance or future citation counts. Journal-specific formatting and submission policies should be checked at submission; the scope of this review is technical strengthening.

REFERENCES / PRIMARY METHODS AND EVALUATION SOURCES

10. Technical references

Author–year citations in the report link to these entries. Titles and source labels below are clickable. For updated preprints, the version date used for this review is stated.

Decoupling compression and routing in attention for time-series forecasting. (n.d.). [Unpublished first draft supplied as *NewTransformer.pdf*, 27 pp.]. Author identities were not specified in the supplied draft.

Dao, T., Fu, D. Y., Ermon, S., Rudra, A., & Ré, C. (2022). FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *Advances in Neural Information Processing Systems, 35*. Paper.

DeepSeek-AI. (2026). *DeepSeek-V4: Towards highly efficient million-token context intelligence* [Preprint]. arXiv. doi:10.48550/arXiv.2606.19348.

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research, 7*, 1–30. Article.

Liu, Y., Hu, T., Zhang, H., Wu, H., Wang, S., Ma, L., & Long, M. (2024). *iTransformer: Inverted transformers are effective for time series forecasting*. International Conference on Learning Representations. Paper.

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Nawrot, P., Li, R., Huang, R., Ruder, S., Marchisio, K., & Ponti, E. M. (2026). *The sparse frontier: Sparse attention trade-offs in transformer LLMs* (Version 3) [Preprint]. arXiv. Original version published in 2025; version reviewed revised June 22, 2026. doi:10.48550/arXiv.2504.17768.

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). *A time series is worth 64 words: Long-term forecasting with transformers*. International Conference on Learning Representations. Paper.

Qiu, X., Hu, J., Zhou, L., Wu, X., Du, J., Zhang, B., Guo, C., Zhou, A., Jensen, C. S., Sheng, Z., & Yang, B. (2024). *TFB: Towards comprehensive and fair benchmarking of time series forecasting methods* [Preprint]. arXiv. Version reviewed: v4, revised August 18, 2025. doi:10.48550/arXiv.2403.20150.

Yuan, J., Gao, H., Dai, D., Luo, J., Zhao, L., Zhang, Z., Xie, Z., Wei, Y. X., Wang, L., Xiao, Z., Wang, Y., Ruan, C., Zhang, M., Liang, W., & Zeng, W. (2025). *Native sparse attention: Hardware-aligned and natively trainable sparse attention* [Preprint]. arXiv. doi:10.48550/arXiv.2502.11089.

Zeng, A., Chen, M., Zhang, L., & Xu, Q. (2023). Are transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*. Paper.

REFERENCES / JOURNAL SELECTION AND RANKING VERIFICATION

11. Journal evidence and final bibliometric check

The Q1 labels below use the latest completed SCImago edition identified during this review: **2025**. Ranking records were checked on **16 October 2026**. Official search-indexed SCImago profiles reported the best-quartile status; direct page retrieval was restricted. No category-specific Scopus CiteScore percentile is presented as independently verified.

Elsevier. (n.d.-a). *What is CiteScore?* Scopus Support Center. Retrieved October 16, 2026. Scopus metric definitions.

Elsevier. (n.d.-b). *Information Sciences* [Journal homepage and aims and scope]. Retrieved October 16, 2026. Journal website.

Elsevier. (n.d.-c). *Neural Networks* [Journal homepage and author information]. Retrieved October 16, 2026. Journal website.

Elsevier. (n.d.-d). *Neurocomputing* [Journal homepage and aims and scope]. Retrieved October 16, 2026. Journal website.

SCImago. (n.d.-a). *Information Sciences* [2025 SJR profile; best quartile Q1; source ID 15134]. SCImago Journal & Country Rank. Retrieved October 16, 2026. Ranking record.

SCImago. (n.d.-b). *Neural Networks* [2025 SJR profile; best quartile Q1; source ID 24804]. SCImago Journal & Country Rank. Retrieved October 16, 2026. Ranking record.

SCImago. (n.d.-c). *Neurocomputing* [2025 SJR profile; best quartile Q1; source ID 24807]. SCImago Journal & Country Rank. Retrieved October 16, 2026. Ranking record.

SCImago. (n.d.-d). *About us*. SCImago Journal & Country Rank. Retrieved October 16, 2026. Data and ranking basis.

Why Neurocomputing is the recommended target

Its published scope includes neurocomputing theory, practice and applications; the revised paper can connect all three through controlled module analysis, a measured implementation and forecasting validation. Information Sciences emphasizes original research in information and intelligent systems, while Neural Networks provides an appropriate stretch audience for a deeper neural-learning contribution. These scope statements support relevance; the prioritization remains a judgment about this particular manuscript. (Elsevier, n.d.-d) (Elsevier, n.d.-b) (Elsevier, n.d.-c)

Record before submission

Archive the journal title and ISSN, current Scopus coverage, required subject category, ranking system, ranking year and category quartile/percentile. Confirm whether the institution accepts SJR best quartile or specifically requires CiteScore Q1 in a named category. Recheck at submission because the relevant ranking year and institutional rules may differ from those used for this report.